

CO3-AT1-Assignment

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QUESTIONS

Q1Banking Transaction System –
Exception Handling

10 marks

**Q2**Hospital Patient Registration –
User-Defined Exception

10 marks

**Q3**E-Commerce Order Processing
– Built-in Exceptions

10 marks

**Q4**Online Food Delivery – Multiple
Threads and Execution Order

10 marks

**Q5**Manufacturing Plant –
Producer-Consumer Problem

Q3 E-Commerce Order Processing – Built-in Exceptions

10 marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 `ArrayIndexOutOfBoundsException` handled Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 `InputMismatchException` handled Invalid input

10 marks

5/5 answered

Your Code

```
1 import java.util.*;
2 class Main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         String[] products={"Laptop","Phone","Headph
6         double[] prices={50000,30000,2000,25000,150
7         try{
8             int index=sc.nextInt();
9             int quantity=sc.nextInt();
10            if(quantity<=0){
11                throw new IllegalArgumentException(
12            }
13            double total=prices[index]*quantity;
14            System.out.println("Product: "+products
15            System.out.println("Total cost: "+total
16        }catch(ArrayIndexOutOfBoundsException e){
17            System.out.println("ArrayIndexOutOfBound
18        }catch(InputMismatchException e){
19            System.out.println("InputMismatchExcept
20        }catch(IllegalArgumentException e){
21            System.out.println("Invalid quantity");
22        }
23        System.out.println("Execution continued.");
24        sc.close();
25    }
26 }
```



✓ Submitted

Input

```
1
2
```

Output

```
Product: Phone
Total cost: 60000.0
Execution continued.
```

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